



DEPARTMENT OF
**DESIGN AND
MERCHANDISING**

LEED + WELL

Aditya Jayadas

**Professor & Graduate Coordinator
Design and Merchandising Department
Oklahoma State University**

LEED

<https://www.usgbc.org/articles/video-what-leed>



Why Green Buildings?



BUILDINGS CONTRIBUTE 39% OF
GLOBAL CARBON EMISSIONS



IMPACT CONSTRUCTION,
OPERATION, AND COMMUNITY
DEVELOPMENT



GREEN DESIGN REDUCES HARM
AND RESTORES NATURAL BALANCE

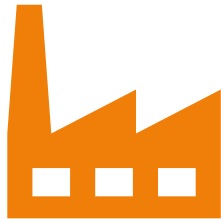
What is LEED?

Leadership in Energy
& Environmental
Design

Voluntary, consensus-
based green building
framework

Certification levels:
Certified, Silver, Gold,
Platinum

History of LEED



1998: LEED v1.0 for New Construction



2007: Green Business Certification Inc. (GBCI) established



2025: LEED v5 ID+C launched

Goals of LEED v5

Decarbonization

Quality of Life

Ecological
Conservation &
Restoration

Benefits of LEED



Lower operating
costs & higher asset
value



Reduced waste to
landfills



Energy & water
conservation



Healthier
environments for
occupants



Tax rebates and
incentives

How LEED is Organized

Eight main credit categories

Prerequisites
(must do) +
Credits (optional)

Point-based
system (0–110
pts total)

LEED Rating Systems



BD+C: Building Design
+ Construction



ID+C: Interior Design +
Construction



O+M: Operations +
Maintenance



Each tailored to
project type

Focus: ID+C (Commercial Interiors)

Applicable to interior renovation projects

Movable furnishings, fixtures, and equipment installed

Scope excludes exterior shell

Minimum Program Requirements

Permanent location on existing land

Reasonable boundaries for project

Size requirements:

- BD+C/O+M: 1,000 sq ft minimum

- ID+C: 250 sq ft minimum

Project Occupancy



REGULAR OCCUPANTS: EMPLOYEES,
RESIDENTS, STUDENTS



VISITORS: CUSTOMERS, PATIENTS,
HIGHER-ED STUDENTS



DEFAULTS PROVIDED BY ASHRAE
62.1-2022

Projects with Incomplete Space

Allowed if <40% gross floor area incomplete

Owner commitment required

Future space must meet credits & prerequisites

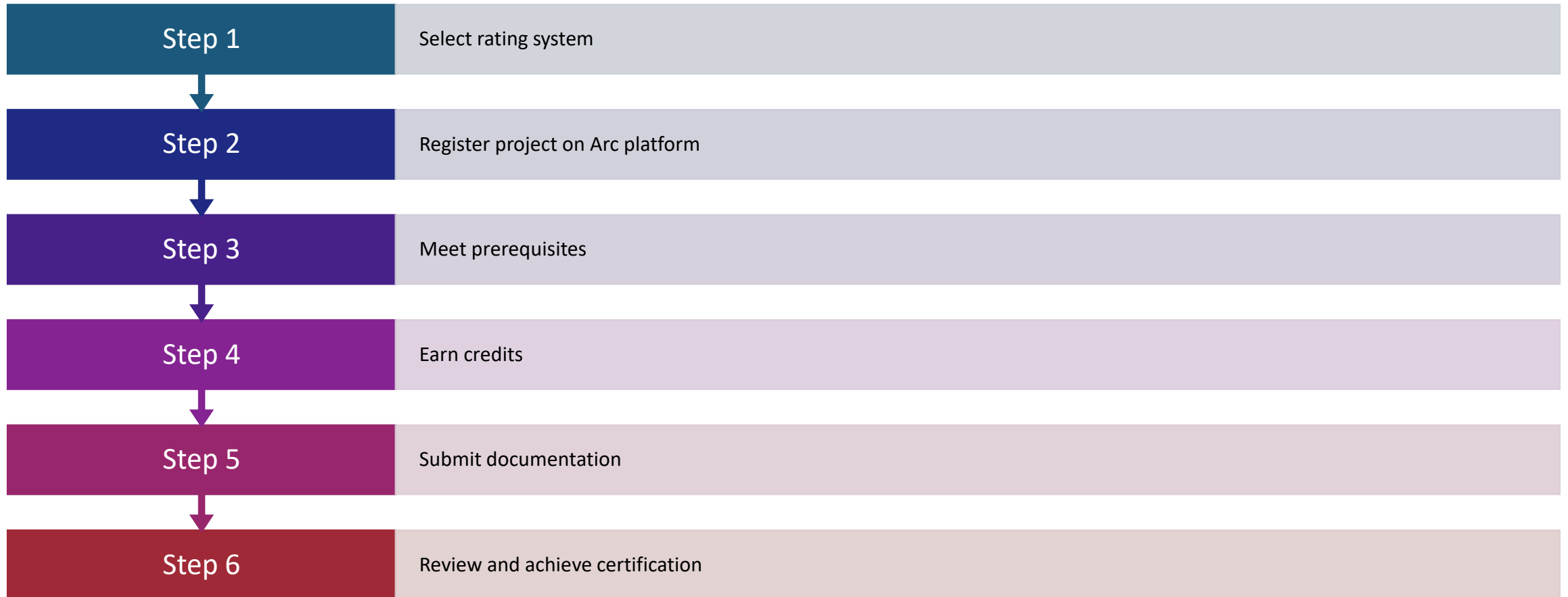
LEED Platinum Requirements

Extra requirements for decarbonization

ID+C: Electrification, Energy Efficiency, Renewables, Embodied Carbon

BD+C and O+M have parallel requirements

LEED Certification Process (Overview)



Documentation Examples



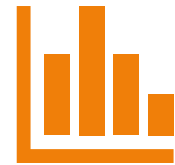
Construction drawings,
specifications, photos



Energy and water
models



Material declarations
and EPDs



Commissioning reports

Certification Levels



Certified: 40–49
points



Silver: 50–59
points



Gold: 60–79
points



Platinum: 80+
points

Review by GBCI



Preliminary Review:
feedback and clarifications



Final Review: decision on
certification level



Appeal process available

Tools and Resources



Arc Platform for
performance tracking



USGBC Glossary



LEED Online submission
portal

Strategies for Success



Plan early with
integrated team



Pursue easy credits
first



Align credits with
project goals



Engage stakeholders
continuously

Integrative Process (IP) – Overview

- Required: Climate Resilience, Human Impact, Carbon Assessment

Integrative Process (IP) – Point Strategies

- Credit: Integrative Design Process (1 pt)

Integrative Process (IP) – Example

- Example: Early workshop including architects, engineers, and tenants

Location & Transportation (LT) – Overview

- Compact Development (8 pts): build in walkable, transit-rich area

Location & Transportation (LT) – Point Strategies

- TDM (4 pts): bike racks, subsidies for public transit

Location & Transportation (LT) – Example

- EV (2 pts): install EV chargers, designate EV parking

Water Efficiency (WE) – Overview

- Minimum Water Efficiency (Req): baseline fixtures

Water Efficiency (WE) – Point Strategies

- Metering & Leak Detection (2 pts): install submeters, leak sensors

Water Efficiency (WE) – Example

- Enhanced Efficiency (8 pts): low-flow fixtures, greywater reuse

Energy & Atmosphere (EA) – Overview

- Electrification (5 pts): replace gas HVAC with electric heat pumps

Energy & Atmosphere (EA) – Point Strategies

- Enhanced Efficiency (12 pts): exceed ASHRAE baseline by 20–30%

Energy & Atmosphere (EA) – Example

- Renewables (5 pts), Commissioning (4 pts),
Grid interactive (3 pts)

Solar Reflectance (SR): The fraction of solar energy reflected by a surface, scored on a scale from 0 (absorbs all heat) to 1.0 (reflects all heat).  Dean Steel Buildings +1

Solar Reflectance Index (SRI): A scale that combines solar reflectance and thermal emissivity (the ability to release absorbed heat), scaled such that a standard black surface is 0 and a standard white surface is 100. Tested per **ASTM**

E1980.  USGBC | U.S. Green Bui... +1

LEED Requirements (LEED v4 / v5 standards)

- **Roofing Materials:**

- Low-slope roofs (2:12 or less) typically require an initial SRI of **82** or a 3-year aged SRI equivalent.
- Steep-slope roofs (greater than 2:12) typically require an initial SRI of **39** or a 3-year aged SRI equivalent.

- **Non-Roof/Hardscape Materials (paving, parking structures):**

- Must have a 3-year aged solar reflectance (SR) value of at least **0.28**.
- If aged data is missing, an initial SR of at least **0.33** at installation is accepted.  USGBC | U.S. Green Bui... +1

- **Cool Exterior Walls (Pilot Credits / LEED v5):**

- Gross exterior wall area can qualify if materials have a solar reflectance of at least **0.60** and thermal emittance of at least **0.75**.  Cool Roof Rating Council

Materials & Resources (MR) – Overview

- Zero Waste + Embodied Carbon (Req)

Materials & Resources (MR) – Point Strategies

- Reuse (4 pts), Low-emitting materials (4 pts)

Materials & Resources (MR) – Example

- Procurement transparency (10 pts), Waste diversion (4 pts)

Indoor Environmental Quality (EQ) – Overview

- Required: Construction Mgmt., IAQ, No Smoking

Indoor Environmental Quality (EQ) – Point Strategies

- Credits: Daylighting, Acoustic quality, IAQ testing

Indoor Environmental Quality (EQ) – Example

- Example: Post-construction air testing for VOCs

Project Priorities (PR) – Overview

- Project Priorities (9 pts): regionally defined sustainability issues

Project Priorities (PR) – Point Strategies

- LEED AP on project team (1 pt)

Project Priorities (PR) – Example

- Example: Stormwater management prioritized in coastal cities

LEED Scorecard – ID+C

- Total: 110 points available
- Categories balanced across sustainability goals
- Choose credits strategically

Example: Small Office Fit-Out

- Enhanced Energy Efficiency (+12 pts)
- Low-Emitting Materials (+4 pts)
- Occupant Experience (+7 pts)
- Total: 65 pts → Gold

Example: Retail Store

- Water Efficiency (+8 pts)
- Transportation Demand Mgmt (+4 pts)
- Reuse Materials (+4 pts)
- Total: 52 pts → Silver

Sector Priorities – Offices

- Focus on Energy, Air Quality, Occupant Experience
- Indoor comfort highly valued

Sector Priorities – Retail

- Focus on Water, Transportation, Materials
- Customer experience considerations

Sector Priorities – Healthcare

- Focus on Air Quality, Accessibility, Resilience
- Patient safety and well-being

Sector Priorities – Education

- Focus on Daylighting, Occupant Experience, Materials
- Improved student performance and attendance

Scoring Challenges

- Some credits overlap with prerequisites
- Documentation burden varies
- Point trade-offs required

How to Maximize Points

- Start with easy credits (low-flow fixtures, no-smoking)
- Pursue high-value credits (energy efficiency, materials transparency)
- Use integrated design process

Market Impacts of LEED

- Global benchmark adopted in 180+ countries
- Linked to property value premiums
- Encourages policy and incentive programs

Challenges of LEED

- Upfront cost perception
- Documentation workload
- Regional adaptation challenges

Future of LEED

- More focus on carbon neutrality
- Integration with smart buildings and AI
- Expanded emphasis on resilience and equity

Key Takeaways

- LEED = framework for measurable sustainability
- ID+C focuses on interiors and tenant fit-outs
- Certification brings environmental and economic benefits

Wrap-Up

- Green buildings = healthier people + healthier planet
- Thank you! Questions?

An aerial, high-angle photograph of a city street intersection at night. The scene is dominated by modern glass skyscrapers with illuminated windows, some showing yellow and blue light patterns. The street below is busy with traffic, creating long, colorful light trails in red, orange, and white. Pedestrian crossings and streetlights are visible. The overall atmosphere is vibrant and urban.

The WELL Building Standard

What is WELL?



A PERFORMANCE-BASED SYSTEM
FOR MEASURING FEATURES OF
THE BUILT ENVIRONMENT



FOCUSES ON HUMAN HEALTH AND
WELL-BEING

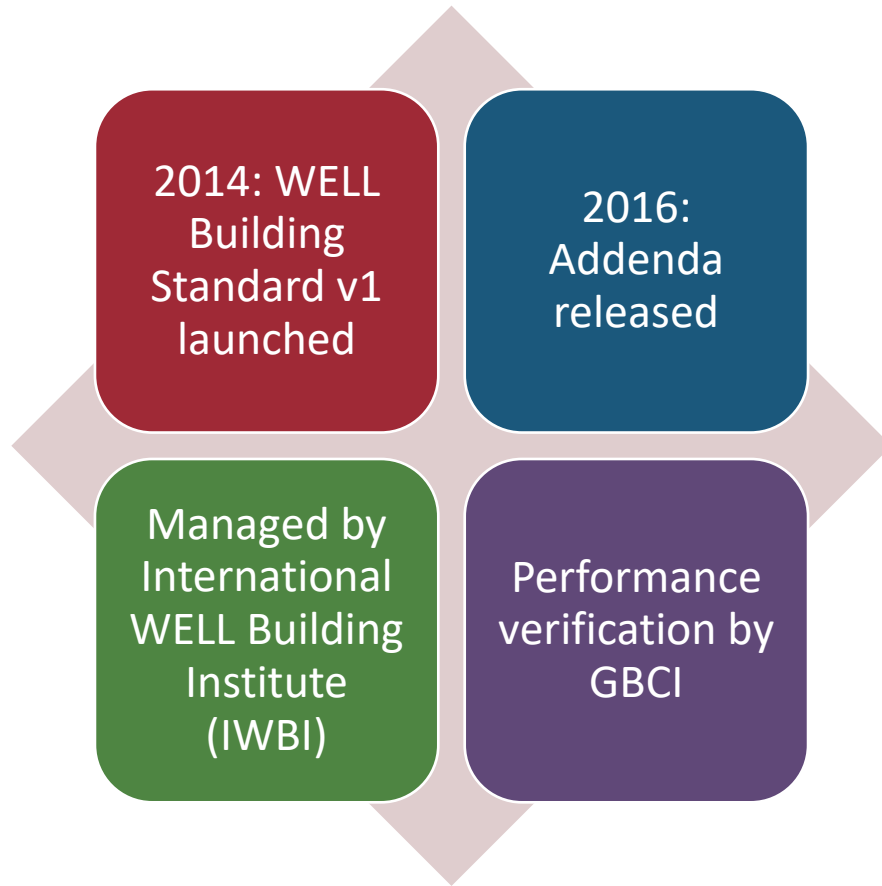


DEVELOPED BY DELOS,
ADMINISTERED BY IWBI



WORKS ALONGSIDE LEED AND
OTHER GREEN BUILDING SYSTEMS

History of WELL



Why WELL?



CENTERS HUMAN HEALTH
AND COMFORT



GOES BEYOND
SUSTAINABILITY →
FOCUSES ON OCCUPANTS



ADDRESSES PHYSICAL,
MENTAL, AND SOCIAL
WELL-BEING



MARKET DEMAND FOR
HEALTHY BUILDINGS

Certification Process Overview



STEP 1: REGISTER PROJECT



STEP 2: IDENTIFY
PRECONDITIONS (MUST-
HAVES)



STEP 3: SELECT
OPTIMIZATIONS

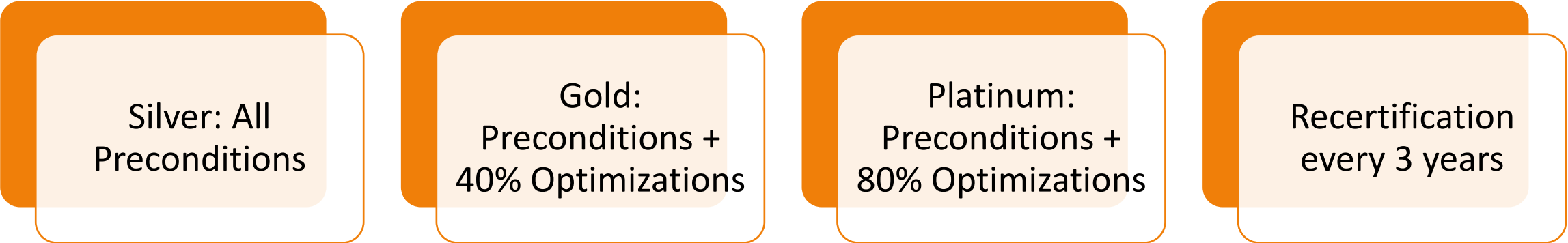


STEP 4: DOCUMENTATION
AND PERFORMANCE
VERIFICATION



STEP 5: CERTIFICATION
AWARDED (SILVER, GOLD,
PLATINUM)

Certification Levels



The diagram consists of four orange rounded rectangular boxes arranged horizontally. Each box contains text describing a certification level. The first box is labeled 'Silver: All Preconditions'. The second box is labeled 'Gold: Preconditions + 40% Optimizations'. The third box is labeled 'Platinum: Preconditions + 80% Optimizations'. The fourth box is labeled 'Recertification every 3 years'.

Silver: All
Preconditions

Gold:
Preconditions +
40% Optimizations

Platinum:
Preconditions +
80% Optimizations

Recertification
every 3 years

Typologies



New and Existing Buildings



New and Existing Interiors



Core and Shell



Pilot Programs: Education,
Retail, Healthcare,
Residential, Hospitality

- Intent: Achieve high indoor air quality

Air Concept – Overview

Air Concept – Preconditions

- Preconditions: Air quality standards, smoking bans, ventilation design

Air Concept – Optimizations

- Optimizations: Enhanced filtration, air quality monitoring, cleaning protocols

Air Concept – Examples

- Examples: High-MERV filters, VOC reduction, IAQ sensors

Water Concept – Overview

- Intent: Provide safe and clean drinking water

Water Concept – Preconditions

- Preconditions: Water quality testing, contaminant limits

Water Concept – Optimizations

- Optimizations: Advanced filtration, water quality promotion



Water Concept – Examples

- Examples: Reverse osmosis systems, hydration stations

Nourishment Concept – Overview

- Intent: Encourage healthy eating habits





Nourishment Concept – Preconditions

- Preconditions:
Fruits/vegetables
availability, food
labeling, allergen
transparency



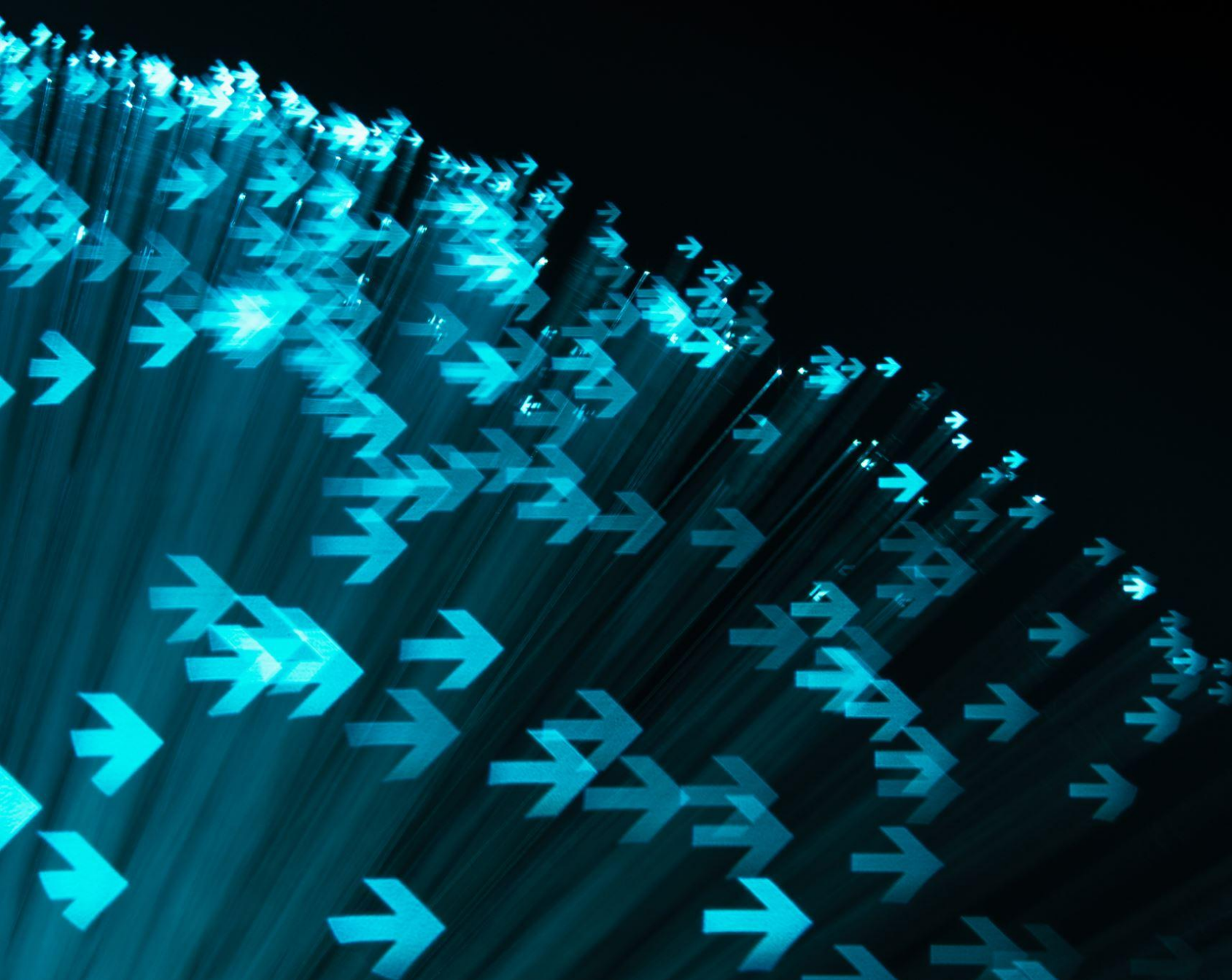
Nourishment Concept – Optimizations

- Optimizations: Portion control, organic produce, mindful eating spaces

Nourishment Concept – Examples

- Examples: Cafeteria offering healthy choices





Light Concept – Overview

- Intent: Support circadian health and productivity



Light Concept — Preconditions

- Preconditions:
Minimum daylight, glare control



Light Concept

—

Optimizations

- Optimizations: Circadian lighting design, daylight simulation



Light Concept – Examples

- Examples: Tunable LED lighting, large windows for daylight



Fitness Concept – Overview

- Intent: Promote physical activity and fitness



Fitness Concept – Preconditions

- Preconditions: Interior circulation, stair access



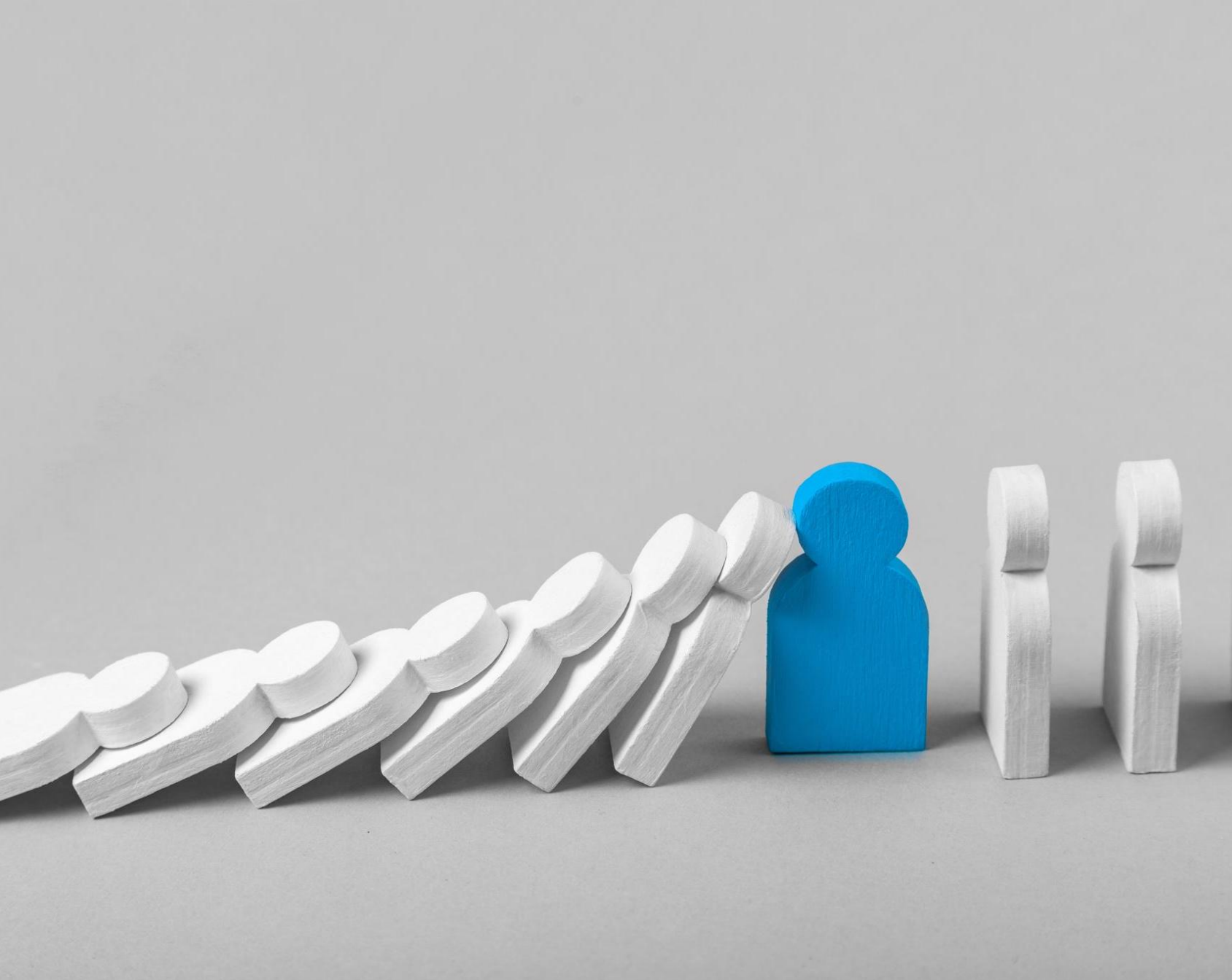
Fitness Concept – Optimizations

- Optimizations: Fitness spaces, active design, ergonomic workstations



Fitness Concept – Examples

- Examples: Central staircases, standing desks, gym facilities



Comfort Concept – Overview

- Intent: Enhance comfort and accessibility

Comfort Concept – Preconditions

- Preconditions: Acoustic control, ergonomic design

Comfort Concept – Optimizations

- Optimizations: Thermal comfort, accessible design features

Comfort Concept – Examples

- Examples: Soundproofing, universal design principles

Mind Concept – Overview

- Intent: Support mental health and well-being

Mind Concept — Preconditions

- Preconditions: Health awareness, biophilic design

Mind Concept — Optimizations

- Optimizations: Stress management programs, sleep support

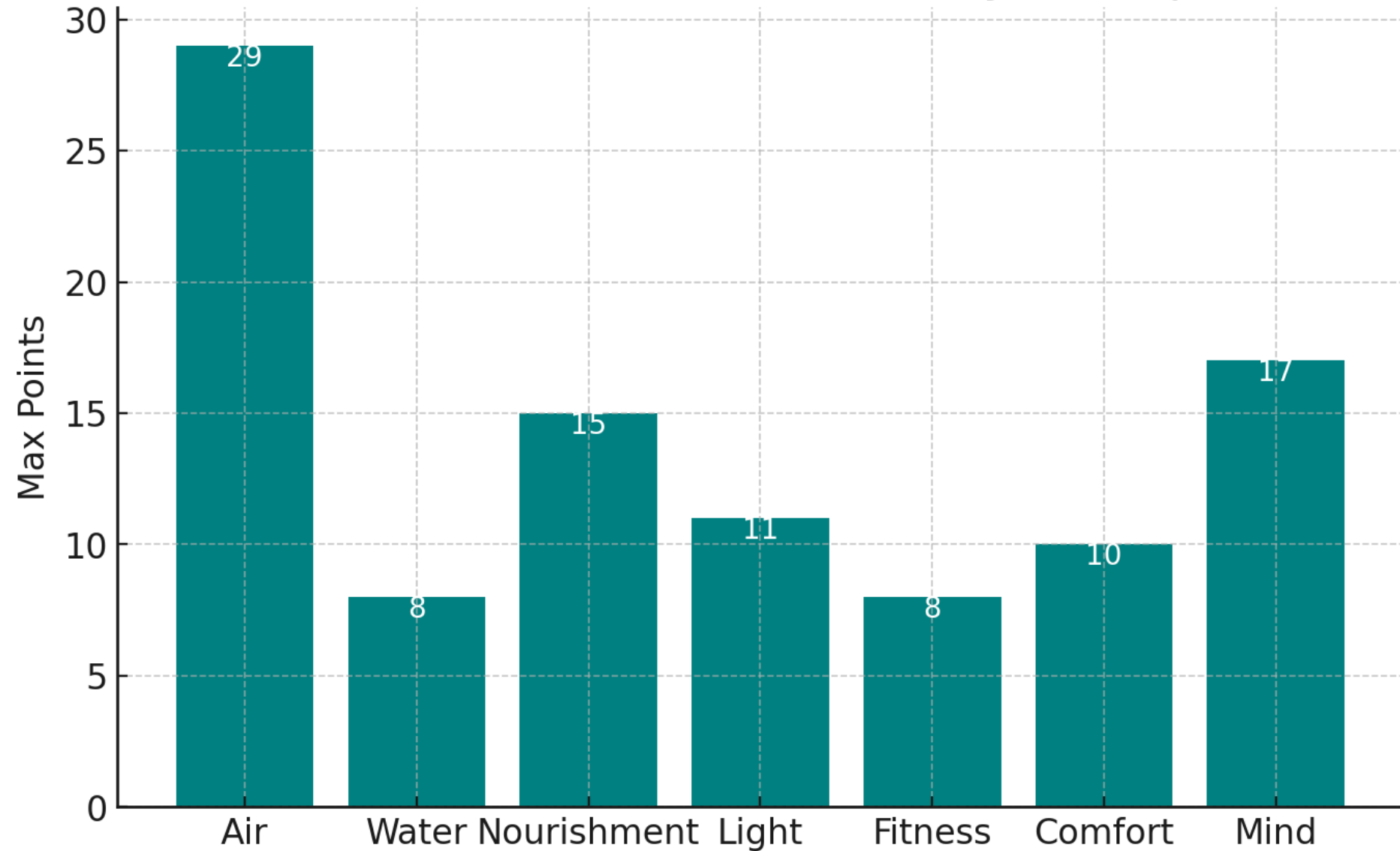
Mind Concept – Examples

- Examples: Indoor greenery, meditation rooms, wellness policies

Scoring WELL

- Preconditions must be met to achieve certification
- Optimizations provide additional recognition
- Higher levels require more optimizations
- Scorecards show compliance across 7 concepts

WELL v1 Points Distribution by Concept



Future of WELL

- Integration with ESG reporting
- Growing demand for healthy workspaces
- AI-driven building management systems
- WELL v2 expansion and updates
- Focus on equity, resilience, and occupant experience

Key Takeaways

- WELL focuses on health and well-being
- Preconditions + Optimizations structure
- 7 Concepts: Air, Water, Nourishment, Light, Fitness, Comfort, Mind
- Certification levels: Silver, Gold, Platinum
- Performance verification critical for success



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